



Analysis of factors that induce cysteine bonding state

Samad Jahandideh^{a,*}, Somayyeh Hoseini^{b,1}, Mina Jahandideh^c, Afsaneh Hoseini^d, Ali Salehzadeh Yazdi^e

^aDepartment of Medical Physics, Shiraz University of Medical Sciences, Shiraz, Iran

^bDepartment of Biochemistry, Division of Genetics, Tabriz University of Medical Sciences, Tabriz, Iran

^cDepartment of Mathematics, Faculty of Science, Vali-E-Asr University, Rafsanjan, Iran

^dDepartment of Computer Engineering, Islamic Azad University of Arak, Arak, Iran

^eInstitute of Biochemistry and Biophysics, University of Tehran, Tehran, Iran

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ABSTRACT

Regarding the fact that the protein structure is principally encoded in its sequence, investigating the bonding state of cysteine has gained a great deal of attention due to its significance in the formation of protein structure. Due to lack of evident influence of free cysteines on the protein structure, it may be expected that only half-cystines convey encoded information. The results obtained from the analysis of amino acid distribution in proximity of both states of cysteines explicitly indicated that prerequisite information for inducing cysteine bonding state is present even in the flanking amino acid sequences of free cysteines.

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1. Introduction

Regarding the structural role of proteins in maintenance of cell shape, and their significance in controlling cellular processes, numerous efforts have been made to investigate the effect of different factors on their function. As a fundamental axiom of biology, the function of a protein is determined by its three-dimensional structure [1], which in turn is resulted at least for small globular proteins only from the protein's amino acid sequence, according to Anfinsen's dogma [2].

Cysteines constitute a special class of amino acids among the amino acid residues. Cysteines are unique, since they can create covalent bonds between two noncontiguous residues in the protein chain [3]. Bonded cysteines, i.e., half-cystines, are among the most influencing factors determining the tertiary structure of the proteins. This significance can be demonstrated by their role in reducing the conformational search space, and stabilizing the folded protein [4]. Therefore, the formation of disulfide bonds between paired cysteine residues is a key step in the folding process of many proteins [5].

Up to now, a large number of investigations have been devoted to the analysis and prediction of disulfide-bonding state of cysteine [6–15]. Regarding the relationship between protein sequence and structure, we scrutinized in the present study the distribution of amino acids in vicinity of both cysteine and cystine forms in order to discover some possible patterns of amino acid occurrence leading to bonded or nonbonded forms of cysteine.

2. Materials and methods

2.1. Database

Because of the availability of more criteria in comparison with similar servers such as PDB_select [16] and PISCES [17], the server PAPIA [18] was used to retrieve the protein sequences satisfying the following criteria. The structures of the desired proteins were determined via X-ray crystallography with resolution higher than 2 Å and *R*-factor lower than 0.2. Protein sequences which either consisted of less than 50 amino acids, were mutant proteins, or had chain break(s), were excluded. No pair of sequences had identity more than 25%. This procedure resulted in 941 protein chains, among which 765 ones had cysteine in their sequences. Taking into consideration the information obtained from DSSP [19] and PDBsum (<http://www.ebi.ac.uk/thornton-srv/databases/cgibin/pdbsum/>) for the remaining 765 chains, the ultimate number of bonded and nonbonded cysteines present in the database were 1067 and 2674, respectively. The high accuracy of the models developed based on the sequence parameters [12,15], leads us to the point that it is

* Correspondence to: Department of Medical Physics, Shiraz University of Medical Sciences, Shiraz, Iran Tel.: +98 71 17266635

E-mail address: jahandideh@sums.ac.ir (S. Jahandideh).

¹ First and second authors have contributed equally to the paper.

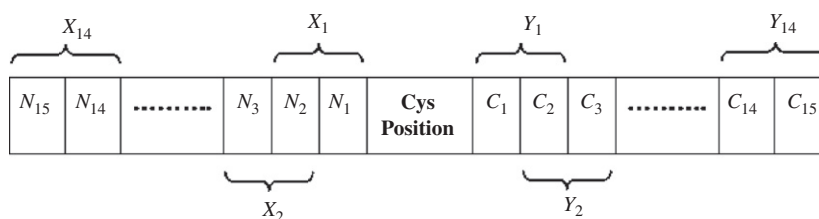


Fig. 1. A schematic representation of defined positions in cysteine-centered windows.

useful to analyze the flanking residues of bonded and nonbonded forms of cysteine. In order to study the distribution of amino acids in proximity of cysteines; we extracted 31-residue cysteine-centered windows.

Considering the mentioned criteria applied for filtering, the results of the analyses carried out on the constructed database are more reliable, compared to those obtained from previous similar databases.

2.2. Definition of positions in cysteine-centered windows

The amino acid positions in the cysteine-centered extracted windows were specified with C_i and N_i , where C and N denote the C-terminal side and N-terminal side of the window, respectively, and i is the position number of the amino acid with respect to the central cysteine. A similar approach was taken to define the double-residue windows over the cysteine-centered windows, in which X_j and Y_j are related to double-residue windows at C-terminal side and N-terminal side of the window, respectively. The notation used is illustrated in Fig. 1.

2.3. Statistical analysis

Test of independence: Due to the presence of conserved regions in protein sequences, the χ^2 test was utilized in order to test the null hypothesis that the two amino acids in a doublet occur independently. The results of the χ^2 test rejected the null hypothesis of independence at a significance level of 0.01 [20].

Local propensities (LPs): The different propensities defined in the present study are singlet local propensity (SLP), observed doublet local propensity (DLP_O), and expected doublet local propensity (DLP_E). In a random distribution of amino acids in different positions, all local propensity values would be 1. Accordingly, an LP value greater than 1 indicate the preference of the amino acid(s) for that position, while an LP value lower than 1 is a sign of avoidance of the amino acid(s) for that position. Based on DLP_O and DLP_E , another propensity called excess doublet local propensity (EDLP) is defined, which indicates the preference of a doublet in the case that its value is positive, while it is the sign of doublet avoidance if its value is negative. The aforementioned LPs are defined as follows.

Singlet local propensity (SLP): We define the SLP of the residue a at the position g_i , where g_i is either N_i or C_i , as

$$SLP(a^{g_i}) = \frac{(n_a^{g_i}/n^{g_i})}{(n_a^{CCW}/n^{CCW})},$$

where $n_a^{g_i}$ is the frequency of amino acid a at g_i position, n^{g_i} is the frequency of all amino acids at g_i position, n_a^{CCW} is the frequency of amino acid a in all cysteine-centered windows, and n^{CCW} is the frequency of all amino acids in all cysteine-centered windows [21].

Observed doublet local propensity (DLP_O): The definition of observed doublet local propensity of the doublet ab at the position h_j ,

where h_j is either X_j or Y_j , is

$$DLP_O(ab^{h_j}) = \frac{(n_{ab}^{h_j}/n^{h_j})}{(n_{ab}^{CCW}/n^{CCW})},$$

where $n_{ab}^{h_j}$ is the frequency of doublet ab at h_j position, n^{h_j} is the frequency of all doublets at h_j position, n_{ab}^{CCW} is the frequency of doublet ab in all cysteine-centered windows, and n^{CCW} is the frequency of all doublets in all cysteine-centered windows.

Expected doublet local propensity (DLP_E): We define the expected doublet local propensity of the doublet ab at the position h_j , where h_j is either X_j or Y_j , as

$$DLP_E(ab^{h_j}) = SLP(a^{g_i}) \times SLP(b^{g_{i+1}}).$$

Excess doublet local propensity (EDLP): In order to obtain an index of preference and avoidance of each doublet, we define Excess doublet local propensity as follows:

$$EDLP = \frac{(DLP_O - DLP_E)}{DLP_E} = \frac{DLP_O}{DLP_E} - 1.$$

3. Results and discussion

3.1. SLP values

The SLP values for each amino acid in 30 positions of the cysteine-centered windows were calculated. The results for bonded and nonbonded 30-aa windows are shown in Table 1a and b, respectively. Since relatively great deviation from SLP value of 1 indicates the preference or avoidance of occurring the amino acid in that specific position for SLP values higher or lower than 1, respectively, these values are highlighted in Table 1a and b. The most preferred amino acids in bonded windows are His, His, Met, Met, Trp and Gln in positions N_{13} , N_{11} , N_{10} , N_8 , N_7 and N_1 , respectively; while the most avoided amino acids in these windows are His, Met, Met, Met, Met and Trp in positions C_{10} , N_7 , C_6 , C_9 , C_{13} , N_3 and N_6 , respectively. On the other hand, Cys, Cys and His in positions N_3 , C_3 and C_4 are most preferred to occur in nonbonded windows.

In addition to analyzing each SLP value separately for each position, it was also investigated how SLP values vary along the cysteine-centered windows. As is shown in Figs. 2a and b, the amino acids Met and Trp exhibit great changes in their SLP values in N-terminal side of bonded windows, especially in positions N_7 – N_8 and N_6 – N_7 , respectively.

In contrast to the asymmetry present in SLP value fluctuation of the other amino acids, an almost perfect symmetry was observed in case of the amino acid cysteine in both bonded and free cysteine-centered windows (Figs. 2a and b). Another surprising finding evidently observed in Figs. 3a and b is that the free cysteine-centered window has a very sharp characteristic signal in positions N_3 and C_3 with SLP values 2.53 and 2.55, respectively. As an astonishing consequence, this clearly points out that, existence of the free-cysteine state depends on the amino acid information content encoded in

Table 1
SLP values of amino acids.

Amino acid	N15	N14	N13	N12	N11	N10	N9	N8	N7	N6	N5	N4	N3	N2	N1	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
A	0.97	1.09	1.12	1.04	0.85	1.08	1.01	0.89	0.88	1.08	1.09	1.07	1.21	1.22	0.98	1.09	1.18	1.09	1.05	0.89	0.97	1.02	0.72	0.99	0.94	0.75	0.77	1.05	0.98	0.91
C	68	77	79	73	60	76	71	63	62	76	77	75	85	86	69	77	83	77	74	63	68	72	51	70	66	53	54	74	69	64
D	0.91	1.02	0.86	0.82	1.12	0.89	0.81	0.97	1.05	1.33	0.99	1.14	0.89	1.07	1.15	1.19	1.25	1.25	0.92	1.04	1.07	1.07	0.95	0.92	0.97	1.00	0.87	0.91	0.74	0.82
E	0.91	0.67	1.19	1.01	0.95	0.97	1.17	0.97	0.91	1.05	0.97	0.75	1.03	1.15	1.01	1.25	1.11	0.73	1.19	1.01	0.97	0.99	1.13	1.05	0.89	1.09	1.09	0.85	0.55	0.50
F	0.93	0.90	0.74	0.95	0.85	0.95	1.11	1.25	1.19	1.03	1.09	1.17	0.87	0.98	0.85	1.01	0.98	0.98	0.87	1.14	0.95	1.19	1.17	1.03	0.90	0.85	0.93	0.82	1.06	1.25
G	0.91	0.97	0.83	0.98	0.93	0.92	1.05	0.84	1.20	1.13	0.99	1.02	0.96	1.25	1.01	1.14	1.13	1.19	1.02	1.02	1.04	0.90	0.89	0.97	1.01	0.90	0.98	0.90	0.99	0.82
H	0.97	0.88	1.42	0.97	1.46	0.93	1.06	1.15	1.15	1.02	0.97	0.80	0.80	0.88	1.11	1.33	0.71	0.88	1.28	0.80	0.93	0.97	1.02	0.97	0.49	1.15	0.88	0.71	1.02	1.28
I	1.15	1.18	0.91	1.18	1.22	0.98	0.86	1.06	1.03	0.94	0.74	1.10	1.06	0.74	0.74	1.01	0.89	0.94	0.94	0.84	0.91	0.98	1.18	1.08	1.10	1.13	1.10	1.06	1.01	0.94
K	0.92	1.09	0.75	1.03	0.96	1.25	1.01	0.96	0.81	1.03	1.03	1.05	1.07	1.03	1.12	0.82	1.05	1.05	1.01	1.31	0.81	0.86	0.92	1.01	0.96	1.14	0.96	1.05	0.86	1.10
L	1.27	0.73	1.18	0.89	0.95	1.00	1.03	1.03	1.06	1.10	0.91	0.86	0.94	1.00	1.06	1.25	1.00	1.10	1.00	0.84	0.84	1.03	0.95	0.97	1.24	0.97	0.89	0.83	1.13	0.95
M	1.12	1.05	1.25	1.38	1.12	1.44	0.85	1.64	0.26	0.72	1.25	1.12	1.12	0.85	1.05	1.12	0.79	0.98	1.44	0.98	0.46	0.92	0.98	0.53	0.98	1.05	1.25	0.59	0.98	0.72
N	1	0.91	0.77	0.93	1.04	0.77	0.83	0.90	0.98	1.11	1.04	1.23	1.11	1.02	1.02	0.93	1.14	1.35	0.93	1.14	1.14	0.81	1.23	0.97	0.88	0.90	1.05	1.09	0.97	0.81
P	0.88	1.17	1.25	1.09	0.84	1.11	1.03	0.98	0.92	0.68	1.05	0.90	1.07	0.94	0.94	1.50	1.01	1.11	0.98	1.11	0.98	1.05	0.84	1.05	0.88	1.11	0.98	0.94	0.84	1.11
Q	0.9	1.05	0.87	0.95	1.00	0.74	1.1	0.77	0.87	1.18	1.08	1.38	1.00	0.82	1.56	1.20	0.92	0.97	0.72	1.31	0.92	1.31	1.00	0.79	0.92	0.90	0.95	0.92	0.72	1.18
R	0.99	0.97	1.04	0.82	1.04	0.99	1.01	1.04	1.04	1.19	0.90	0.77	1.39	1.08	1.17	1.15	1.15	0.66	0.99	1.12	1.34	1.10	0.62	1.06	0.93	0.93	0.86	0.97	0.77	0.93
S	0.92	0.81	0.89	1.08	0.95	0.89	0.94	1.10	1.15	0.91	1.09	1.18	1.24	1.05	1.33	1.00	1.02	0.86	1.20	1.01	1.08	0.96	0.94	1.09	0.80	0.92	0.99	0.92	0.96	0.73
T	0.9	0.94	0.93	0.93	1.08	0.97	1.10	0.94	0.98	1.07	1.07	1.00	1.11	1.24	1.15	0.90	1.12	1.22	1.00	0.89	1.17	0.90	1.01	1.00	1.00	0.8	0.96	1.00	0.98	0.89
V	0.83	1.21	0.82	1.05	0.98	1.06	1.11	1.01	0.90	0.87	1.18	0.95	1.10	1.03	1.06	0.95	0.96	1.03	1.16	0.87	0.93	0.88	1.18	0.83	0.98	1.21	1.01	0.96	0.90	0.98
W	1.31	1.03	1.31	1.31	1.03	1.14	0.69	1.20	1.49	0.40	1.03	1.20	0.51	0.74	1.03	0.74	0.91	0.74	0.86	0.97	1.20	0.91	1.26	1.14	1.14	1.03	0.80	0.80	1.20	0.86
Y	0.9	0.94	1.13	0.78	1.10	0.99	1.13	1.2	1.04	0.97	1.10	1.29	0.87	1.20	0.97	0.67	0.85	0.85	1.04	0.92	0.92	1.27	0.81	0.87	0.92	1.04	1.06	1.10	0.99	1.10
	39	41	49	34	48	43	49	52	45	42	48	56	38	52	42	29	37	37	45	40	40	55	35	38	40	45	46	48	43	48

(a) Bonded cysteine-centered windows

Table 1 (Continued).

Amino acid	N15	N14	N13	N12	N11	N10	N9	N8	N7	N6	N5	N4	N3	N2	N1	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	SLP	
A	0.96	1.02	0.99	0.97	1.10	1.04	0.87	0.95	0.90	0.91	0.96	0.92	0.96	1.08	1.09	0.88	1.10	0.98	1.01	1.05	1.04	1.11	1.06	0.98	1.02	1.03	1.02	0.97	1.07	0.96
	197	210	203	199	226	215	179	196	185	188	198	190	198	222	224	182	227	202	208	217	214	228	218	201	211	212	210	200	221	197
C	0.87	0.88	0.95	0.71	0.99	0.83	0.83	0.88	1.16	1.02	0.92	1.04	2.53	0.61	0.82	0.82	0.66	2.55	1.02	0.95	1.02	1.14	0.83	0.83	0.83	0.99	0.73	0.93	0.80	0.87
	51	52	56	42	58	49	49	52	68	60	54	61	149	36	48	48	39	150	60	56	60	67	49	49	49	58	43	55	47	51
D	1.03	1.01	1.11	1.04	1.02	0.94	1.04	0.99	1.13	1.03	1.07	0.95	0.91	1.02	0.83	1.02	1.10	0.96	0.97	1.11	1.18	0.90	0.99	1.05	0.98	0.85	0.98	0.96	0.87	0.94
	161	159	175	164	160	148	164	156	178	161	168	149	143	160	130	160	172	151	153	175	185	142	155	165	154	134	154	151	137	147
E	0.95	0.98	1.11	0.96	0.91	1.10	1.13	0.99	0.94	1.13	0.94	0.78	0.96	1.07	0.92	0.86	0.90	0.84	1.03	1.06	1.18	0.90	1.01	1.05	1.03	1.03	0.95	1.06	1.16	1.06
	158	163	184	159	151	183	188	165	156	187	156	130	160	177	153	142	150	140	170	176	195	150	167	174	170	170	158	176	192	175
F	0.90	0.93	0.91	0.91	0.97	1.02	0.94	1.11	0.90	1.03	0.98	1.24	1.03	1.14	1.04	0.98	0.86	1.07	0.97	0.86	0.96	1.11	0.99	0.75	1.14	1.06	1.11	0.91	1.11	1.05
	98	101	99	99	105	111	102	120	97	112	106	134	112	124	113	106	93	116	105	93	104	120	107	81	124	115	120	99	120	114
G	0.96	0.86	0.82	1.01	1.01	0.88	1.01	1.08	1.08	1.04	1.10	0.94	1.02	1.05	1.13	1.16	1.08	0.95	1.21	0.95	0.93	1.03	1.10	0.94	0.95	0.95	0.89	1.02	0.89	0.98
	204	182	174	215	215	186	214	229	230	221	234	200	217	222	239	246	229	201	257	202	197	218	234	199	201	202	190	217	189	208
H	0.93	0.87	1.01	1.15	1.07	0.99	0.95	1.01	1.10	1.15	0.93	0.95	0.99	0.70	1.16	1.64	1.10	1.06	1.46	1.00	0.92	1.00	0.87	0.91	0.89	0.62	0.93	0.99	0.76	0.91
	69	64	75	85	79	73	70	75	81	85	69	70	73	52	86	121	81	78	108	74	68	74	64	67	66	46	69	73	56	67
I	0.99	0.89	1.11	0.91	0.97	0.85	1.04	1.10	0.93	0.91	1.12	1.21	1.08	1.14	1.00	1.15	0.91	1.04	0.90	0.92	0.81	0.96	0.99	1.12	0.95	0.93	0.96	0.93	1.06	1.12
	143	129	161	132	141	123	151	159	135	132	163	176	157	165	145	166	132	150	130	133	117	139	143	163	138	135	139	135	153	162
K	1.13	1.06	1.06	0.91	1.00	1.15	1.08	0.95	1.02	1.11	0.86	0.86	0.75	0.87	0.88	0.82	0.97	1.00	0.83	0.88	1.16	1.06	0.95	1.22	1.21	0.97	1.11	0.99	1.08	1.05
	164	154	154	131	145	166	156	138	147	160	125	124	109	126	128	118	141	144	120	128	168	153	137	176	175	141	160	143	156	152
L	1.06	1.07	0.99	0.98	1.02	1.04	0.93	1.02	1.01	0.80	1.05	1.24	1.05	1.00	1.05	1.02	1.02	1.06	1.06	0.92	0.82	0.92	0.97	1.02	1.05	0.98	0.99	0.95	0.87	1.02
	243	246	228	226	235	239	213	235	231	183	242	285	242	230	240	235	234	243	244	212	188	210	222	233	240	226	227	217	200	235
M	0.89	1.09	0.93	1.00	1.00	1.00	1.09	0.92	1.27	1.12	1.15	1.22	1.03	1.12	0.79	0.96	0.89	1.09	1.22	1.30	1.01	1.09	1.03	0.85	0.84	1.03	0.79	0.73	0.81	0.76
	56	69	59	63	63	63	69	58	80	71	73	77	65	71	50	61	56	69	77	82	64	69	65	54	53	65	50	46	51	48
N	1.06	0.90	0.92	1.05	0.95	1.12	1.24	1.09	1.04	1.00	1.12	1.06	1.04	0.98	1.16	1.03	1.16	1.03	0.96	1.12	1.09	0.98	1.01	1.07	0.76	0.88	1.02	0.77	0.86	0.91
	117	100	102	116	105	124	137	121	115	111	124	117	115	108	128	71	128	114	106	124	120	108	112	118	84	97	113	85	95	101
P	0.89	0.83	1.12	0.96	0.93	0.98	1.01	1.02	1.10	1.02	0.99	0.81	0.86	0.82	0.87	0.99	0.98	0.93	1.00	1.19	1.03	1.20	1.06	1.05	1.15	1.15	0.99	1.04	0.99	1.04
	112	104	141	120	116	123	127	128	138	128	124	102	108	103	109	124	123	116	125	149	129	150	133	132	144	144	124	131	124	131
Q	1.08	1.09	1.01	0.99	0.94	1.03	1.16	1.00	1.18	1.17	0.86	0.83	0.95	0.97	1.07	1.24	1.00	0.78	0.81	0.98	1.05	0.83	0.95	0.99	1.14	0.89	0.99	1.11	1.06	0.88
	106	107	99	97	92	101	114	98	116	115	84	81	93	95	105	122	98	77	79	96	103	81	93	97	112	87	97	109	104	86
R	0.92	1.06	1.04	1.05	0.93	1.01	0.95	0.94	0.95	1.10	1.09	0.98	0.82	0.93	1.02	1.17	1.03	0.86	1.01	1.09	1.07	0.80	1.02	1.16	0.97	0.90	0.96	1.17	1.01	0.95
	115	133	130	132	116	127	119	118	119	138	137	123	103	117	128	147	129	108	127	137	134	100	128	145	122	113	120	146	127	119
S	1.04	1.07	1.04	0.99	1.10	0.99	0.97	1.02	0.86	1.03	0.99	1.11	0.93	0.89	1.15	1.12	1.21	0.96	0.98	0.95	1.05	1.11	0.97	0.91	0.95	1.06	0.95	0.78	0.89	0.95
	155	160	155	147	164	148	144	152	128	153	147	166	138	132	171	167	181	143	146	142	157	165	145	136	141	158	142	117	132	141
T	0.95	1.07	0.88	1.09	0.97	1.10	1.11	0.98	0.93	1.11	1.01	0.93	1.06	1.15	1.02	1.01	0.87	0.98	0.89	1.00	0.92	1.03	1.00	0.99	0.88	1.02	1.06	0.94	1.04	1.00
	137	154	126	157	139	158	160	140	133	159	145	134	152	165	146	145	125	141	128	143	132	148	144	142	127	147	152	135	149	144
V	0.91	0.99	0.89	1.09	0.94	0.82	0.98	0.94	0.96	1.03	1.14	1.12	1.19	1.18	1.01	0.93	1.01	0.93	0.99	1.06	0.91	1.04	1.02	0.89	0.91	1.04	1.04	1.17	0.94	0.93
	165	179	162	197	170	149	177	170	173	186	207	203	215	214	182	169	183	169	179	192	164	188	185	161	165	188	189	212	171	168
W	1.19	0.78	0.91	0.78	0.78	1.06	0.62	1.01	1.14	0.96	0.70	1.11	0.88	1.06	1.22	0.83	0.98	1.14	1.14	0.78	1.34	0.98	0.93	1.01	0.67	1.27	1.24	1.11	1.22	1.19
	46	30	35	30	41	24	39	44	37	43	37	43	34	41	47	32	38	44	44	30	52	38	36	39	26	49	48	43	47	46
Y	0.90	1.01	0.85	1.03	1.13	1.03	0.78	0.96	0.98	0.69	0.85	1.08	0.91	1.16	1.05	1.15	1.14	1.09	0.96	0.96	1.02	0.98	1.00	0.96	1.13	1.21	0.97	1.00	1.15	0.86
	84	94	79	96	105	96	72	89	91	64	79	100	85	108	98	107	106	101	89	89	95	91	93	89	105	112	90	93	107	80

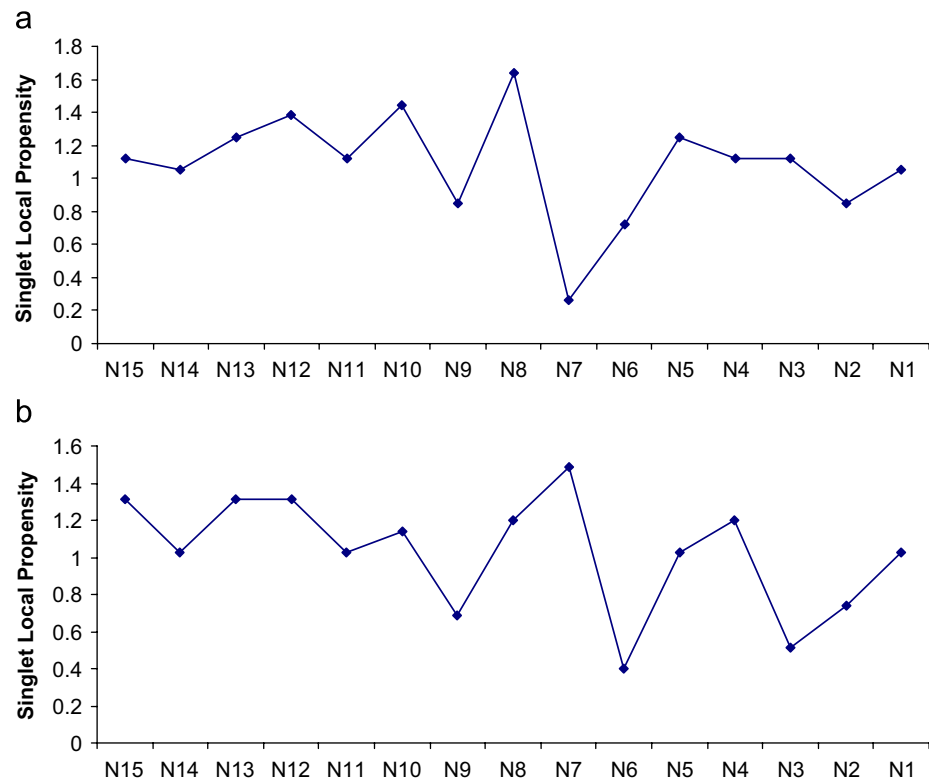


Fig. 2. SLP changes of (a) Met and (b) Trp in N-terminal side of bonded cysteine-centered windows.

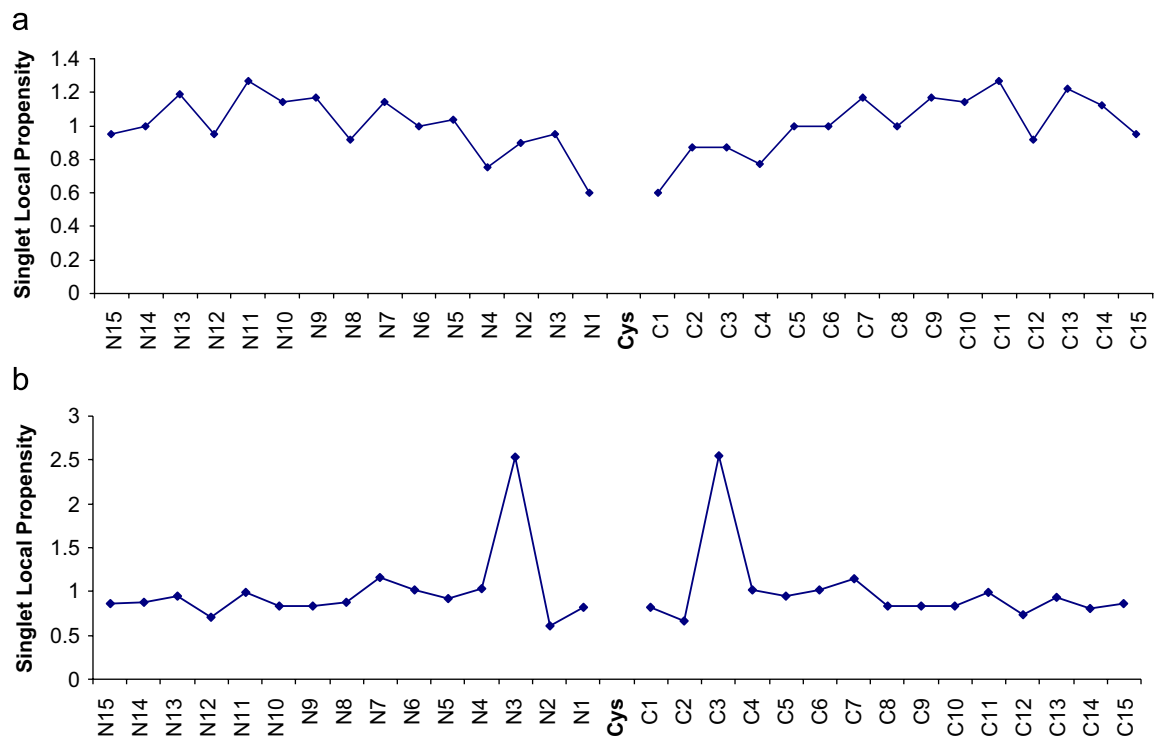


Fig. 3. SLP changes of the amino acid cysteine in (a) bonded and (b) nonbonded cysteine-centered windows.

the flanking amino acid sequences of the free cysteines. Existence of characteristic signals is consistent with the general idea that the information required for the formation of protein structure is

encoded in the protein sequence. Such an approach has been already applied on α -helices, resulting in the conclusion that amino acid propensities are position-dependent throughout the length of this

Table 1
EDLP values for exceptional doublets.

X14 Doublet EDLP	X13 Doublet EDLP	X12 Doublet EDLP	X11 Doublet EDLP	X10 Doublet EDLP	X9 Doublet EDLP	X8 Doublet EDLP	X7 Doublet EDLP	X6 Doublet EDLP	X5 Doublet EDLP	X4 Doublet EDLP	X3 Doublet EDLP	X2 Doublet EDLP	X1 Doublet EDLP
<i>(a) N-terminal side of bonded cysteine-centered windows</i>													
Met-Pro 2.7260	His-Leu 3.6123	Gly-His 2.3214	Met-Met 4.2001	Phe-Trp 7.6669	Met-Asn 3.0820	Ala-Cys 2.9560	His-Met 7.4263	Pro-Trp 7.0321	Ile-Ile 2.1535	Ala-Trp 2.5556	His-Cys 3.3619	Ala-Ala 0.8757	Thr-Phe 2.2972
Ile-Phe 2.6452	Arg-Ile 2.8762	Asp-Trp 1.9273	His-Ile 2.7105	Gln-Gln 1.5146	Tyr-Arg 1.6170	His-Lys 2.8079	Gln-Asp 3.3215	Trp-Val 2.1985	Gln-Glu 2.0568	Val-Gly 1.3967	Arg-Cys 2.3490	Asp-Thr 1.1953	Met-Gln 2.1121
Tyr-Asn 2.4147	Tyr-Phe 2.7213	Arg-Val 1.8141	Val-Trp 1.7386	Tyr-Trp 1.2186	Asp-Gly 1.1613	Gly-Gln 2.6644	His-Pro 2.4233	Lys-Gly 0.4260	Pro-Pro 1.9165	Ser-Tyr 1.0841	His-Ile 1.6283	Phe-His 1.4532	Arg-His 1.8607
Ile-Asp 1.6092	His-Gly 2.1664	Ile-Pro 1.4889	Trp-Arg 1.3502	Cys-Pro 0.8368	Tyr-Ile 1.5999	Trp-Gln 2.4277	Tyr-Trp 1.7569		Ile-Asn 1.3712		Lys-Ser 1.4332	Phe-Arg 1.9712	Asn-Glu 1.8407
Arg-Met 1.1791	Arg-Pro 1.1862	Ser-Leu 1.0507		Ser-Cys 0.8255	Cys-Ala 1.2836	Tyr-Phe 2.2462	Asn-Ile 1.5581		Val-Cys 1.3491			His-Asp 2.1492	Pro-Pro 1.5478
Asp-Val 0.9361	Glu-Met 1.8542			Gly-Ser 0.6312	Ser-Cys 1.1536	Phe-Asn 1.8541	Glu-Tyr 1.5306		Gln-Thr 1.2808			Met-Asp 1.2636	Thr-Asn 0.6458
Ser-Gly 0.5602	Thr-Val 1.7275					Glu-Asn 1.2673	Val-Asp 1.2685		Gly-Gly 0.8613			Thr-Arg 1.3924	Trp-Gln 0.6260
	Phe-Ser 1.6127						Glu-Cys 1.2554					Thr-Tyr 1.4669	
							Leu-Arg 1.0738					Val-Asn 1.0530	
												Trp-Ala 1.1488	
Y14 Doublet EDLP	Y13 Doublet EDLP	Y12 Doublet EDLP	Y11 Doublet EDLP	Y10 Doublet EDLP	Y9 Doublet EDLP	Y8 Doublet EDLP	Y7 Doublet EDLP	Y6 Doublet EDLP	Y5 Doublet EDLP	Y4 Doublet EDLP	Y3 Doublet EDLP	Y2 Doublet EDLP	Y1 Doublet EDLP
<i>(b) C-terminal side of bonded cysteine-centered windows</i>													
Gln-Trp 3.8792	Met-Trp 8.1140	Arg-Arg 3.6643	Met-Asn 3.3967	Trp-Phe 3.3334	Ala-Asn 1.3738	Glu-Asn 0.9343	Trp-Arg 3.2531	Trp-His 4.9924	Val-Asp 1.7593	His-Lys 2.4440	Met-Ile 4.3612	Gln-His 3.9009	Asn-Ile 1.8607
Met-Thr 2.5674	Val-Gln 1.8102	Glu-His 2.7196	Phe-Cys 3.0997	Ile-His 2.8340	Phe-Glu 1.8643	His-Arg 3.7279	Leu-His 1.7196	Tyr-Pro 2.5542	Pro-Glu 1.0605	Met-Arg 1.7594	Lys-Phe 3.2742	Thr-Met 2.8898	His-Asn 1.8407
His-Leu 2.3315	Asn-Leu 1.4104	Trp-Gly 2.1738	Pro-Phe 2.5213	Trp-Asp 2.1475	Gln-Asn 2.2886	Met-Glu 2.3823	Gln-Gly 1.5600	Ile-Ile 2.0589	Ala-Ala 0.6784	Asn-Lys 1.3641	Tyr-Tyr 1.7390	Cys-Arg 2.2972	Leu-Asp 1.5478
Gly-Ser 2.0631	Thr-Asn 1.3405	Cys-Ala 1.8946	Trp-Ala 2.1546	Pro-Lys 2.1474	Thr-Met 1.9219	Asn-Trp 1.6912	Asn-Thr 1.4441	Pro-Val 1.9544		Pro-Glu 1.5793	Gly-Lys 0.5883	Gln-Gln 2.1185	
Asn-Arg 1.9799	Arg-Val 1.3170	Glu-Phe 1.5616	His-Gly 1.9004	Gly-Tyr 1.4278		Pro-Thr 1.3032	Gln-Trp 1.4352	His-Leu 1.8068		Gln-Tyr 1.2034		Trp-Gln 2.0443	
Trp-Val 1.7985	Asp-Gly 1.0701	Asn-Asp 1.0576	Gln-Ile 1.6190			Gln-Gln 1.9405	Val-Cys 1.4209	Asn-Asn 1.5532		Trp-Val 3.4353		Trp-Asn 1.8288	
Arg-Met 1.7168	Cys-Pro 1.0274	Gln-Tyr 0.7369	Cys-Arg 1.6109			Ser-Met 4.3384	Arg-Met 1.0919	Arg-Met 1.4496		Tyr-Ile 1.5554		Asn-Pro 1.1552	
Leu-Gln 1.6339	Lys-Glu 1.0055		Tyr-Trp 1.5356			Thr-Phe 0.8015	Ala-Ser 0.8371	Asn-Gln 1.2070				Asp-Val 0.6458	
Asn-Ile 1.6153	Ser-Phe 0.7406		Asn-Asp 1.1981					Asp-Val 0.7960				Gly-Ser 0.6260	
Glu-Ile 1.1299	Ala-Ala 0.6898		Tyr-Ser 1.0682										
Cys-Pro 0.4572	Gly-Ser 0.5588		Ile-Val 0.9933										
			Arg-Met 0.9661										

regular and repetitive structure [22]. Based upon the results of the mentioned studies, the viewpoint can be extended to other structural elements in order to decipher information hidden in the protein sequences.

3.2. Exceptional doublets

Though singlet local propensity provides valuable information on the occurrence of amino acids in single position, it may miss some sequence-order information. Doublet local propensity contains further information on the simultaneous occurrence of two amino acids in adjacent positions. In order to check the interdependence of amino acids present in each doublet, the chi-square (χ^2) test was applied. Out of 11 200 possible doublets, i.e., 400 possible doublets in each

double position multiplied by 28 doublet positions, only for 168 and 187 doublets in nonbonded and bonded cysteine-centered windows, the null hypothesis of independence was rejected at significance level of 0.01. These exceptional doublets are listed in Tables 2a and b and 3a and b.

Definition of EDLP is based upon DLP_E and DLP_O , whose values indicate the expected and observed doublet LPs of amino acids. The absolute value of EDLP offers a measure of the preference or avoidance of the doublet at that specific position, while its positive or negative sign specifies whether its occurrence is preferred or avoided, respectively.

For the exceptional doublets mentioned above, all but five doublets were preferred in nonbonded cysteine-centered windows, whereas all exceptional doublets were preferred in bonded cysteine-centered windows (Tables 2a and b and 3a and b).

Table 2
EDLP values for exceptional doublets.

X14 Doublet EDLP	X13 Doublet EDLP	X12 Doublet EDLP	X11 Doublet EDLP	X10 Doublet EDLP	X9 Doublet EDLP	X8 Doublet EDLP	X7 Doublet EDLP	X6 Doublet EDLP	X5 Doublet EDLP	X4 Doublet EDLP	X3 Doublet EDLP	X2 Doublet EDLP	X1 Doublet EDLP
<i>(a) N-terminal side of nonbonded cysteine-centered windows</i>													
Trp-Gln	Gln-His	Trp-Cys	Cys-Phe	Cys-His	Ser-His	Asp-Trp	Cys-Gln	Val-Ser	Pro-Asp	Lys-His	Trp-Ser	Glu-Pro	Trp-Lys
2.0120	2.1643	2.5090	2.7960	0.6514	1.9145	1.6918	2.3519	0.9362	1.2980	1.7104	1.8422	1.3896	1.3735
Met-Met	Asp-Tyr	His-Met	Cys-Tyr	Asp-Glu	Tyr-Asn	Asn-Trp	Phe-Trp	Trp-Lys	Cys-Ile	Asn-Arg	Cys-Val	Ser-Cys	ASN-Pro
1.6828	1.2081	2.3923	1.9933	0.7462	1.2062	1.0362	2.1734	0.7693	1.2823	1.2423	1.7037	1.2485	0.8750
Tyr-Tyr	Cys-Gly	Met-Arg	Ile-Phe	Asp-Ile	Glu-Asp	Arg-Val	Ala-Lys	His-Pro	Glu-Ala	His-Pro	Lys-Cys	Val-Asn	
1.1290	0.8523	1.5307	1.7777	0.7039	1.0490	0.6013	1.1777	0.5649	0.5192	1.0492	1.6132	0.9725	
Ile-Ser		Cys-Thr	Val-Glu	Phe-His	Pro-Phe	Met-Ala	Ser-Ala	Arg-Val		Phe-Asp	Gln-Cys	Val-Lys	
1.0338		1.3153	1.3723	1.8897	1.0461	0.8701	0.8040	0.5611		0.8850	1.2031	0.7673	
Val-Glu		Asn-Trp	Ser-Cys	Leu-Gln	Gly-Lys		Val-Leu	Cys-His		Lys-Lys	Val-Lys	Leu-Gly	
0.9538		1.3702	1.1837	0.7938	0.7285		0.6423	0.0369		0.8139	1.1637	<u>−0.5639</u>	
Pro-Glu		Pro-Gln	His-Pro	Tyr-Phe	Phe-Asn		Leu-Thr			Met-Thr	Trp-Arg		
0.9504		1.0861	0.6455	0.7965	0.6232		0.6032			0.7071	1.5943		
Met-Thr		Gly-Ser								Asp-Gly			
0.9363		0.8087								0.6691			
His-His		His-Pro								Ala-Cys			
0.7158		0.8027								0.6245			
Thr-Val		Cys-His								Val-Val			
0.6989		0.6788								0.4947			
Y14 Doublet EDLP	Y13 Doublet EDLP	Y12 Doublet EDLP	Y11 Doublet EDLP	Y10 Doublet EDLP	Y9 Doublet EDLP	Y8 Doublet EDLP	Y7 Doublet EDLP	Y6 Doublet EDLP	Y5 Doublet EDLP	Y4 Doublet EDLP	Y3 Doublet EDLP	Y2 Doublet EDLP	Y1 Doublet EDLP
<i>(b) C-terminal side of nonbonded cysteine-centered windows^a</i>													
Asn-Glu	His-Tyr	Arg-Trp	Tyr-Val	Tyr-Thr		Asp-Gly	Ser-Gln	Tyr-Trp	Ser-Cys	Thr-Thr	Arg-Val	Tyr-Ile	Cys-His
1.7382	1.9384	2.4585	1.7346	1.7735		1.3343	1.6236	7.8281	1.5198	3.5191	4.4766	2.5232	1.0992
Phe-Asp	Pro-Glu	Cys-His	Tyr-Val	His-Phe		Gln-Tyr	Lys-Glu	Thr-Gly	Met-Thr	Trp-Tyr	Tyr-Arg	Thr-Leu	Val-Phe
0.7585	1.3110	1.8363	0.9099	1.6226		1.0948	1.4550	3.4020	0.9557	1.8646	1.3071	2.0934	1.5249
His-His	Asp-Ser	Met-Arg	His-His	Arg-His		Tyr-Phe	Trp-Ser	Tyr-Asp	Lys-Asn	Trp-Asn	Trp-Trp	Glu-Ala	Leu-Lys
0.3915	1.0338	1.5303	0.7065	1.1591		0.6427	1.3647	3.0781	0.8938	1.6569	1.0509	1.9363	1.3018
Pro-Asp	Glu-Ala	His-Gly	Cys-Gly	Tyr-Phe		Cys-Gln	Gln-Leu	Cys-Arg	Cys-Trp	His-His	His-His	Tyr-Cys	His-Ile
<u>−1.0000</u>	0.6422	0.9178	0.6901	0.8809		0.6325	2.2050	2.2050	0.8076	1.4634	0.8907	1.8386	1.2879
	Ser-Glu	Gly-Cys	Gly-Glu	Val-Leu			Asn-Phe	Gln-Pro		Leu-Gln	Pro-Phe	Glu-Trp	Asn-Val
	<u>−0.8924</u>	0.6780	<u>−0.5999</u>	0.8554			0.8717	1.4691		1.3894	0.6179	1.4068	1.2693
		Asn-Pro		Pro-Val				Pro-Trp		Cys-Pro		Lys-Lys	Ile-Gly
		0.5489		0.8241				1.1743		1.1464		1.1218	0.9508
		Leu-Ala		Cys-His				Trp-Met		Phe-Glu		Leu-Ser	Asn-Trp
		0.5056		0.3115				0.9929		0.9965		1.1076	0.9238
		Ala-Ala						Arg-Trp		Arg-His		His-His	Glu-Lys
		0.4131						0.5478		0.8338		1.0480	0.7038
		Cys-Asp								His-Ile		Pro-Glu	Trp-Arg
		<u>−1.0000</u>								0.5535		0.8068	0.5860
												Met-Ala	Cys-Asp
												0.5356	0.4478
												Cys-Ile	
												0.4377	

^aBold, italic, and underlined numbers show negative EDLP values.

4. Conclusion

Analysis of amino acid distribution in neighborhood of both bonded and nonbonded forms of cysteine was motivated by the idea of developing a more informative framework for protein design. The results unexpectedly indicated that the flanking sequences of free cysteines contain essential information for inducing the free form of cysteine. Being used as a signal, it can propose a novel feature to predict the bonding state of cysteine.

For future studies, it is recommended that further information hidden in distribution of amino acids would be extracted in different structural and functional levels, in order to find more specific signals useful for protein design.

Conflict of interest statement

The authors declare that they have no competing interests.

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Samad Jahandideh graduated from University of Shahid Chamran, Iran, with a B.Sc. degree in Genetics in 2003. He received his M.Sc. degree in Biophysics in 2005 from Tarbiat Modares University. He is pursuing his Ph.D. degree at Tarbiat Modares University in Biophysics.

Somayyeh Hoseini holds a B.S. degree in Biology from University of Arak, Iran, in 2004. She is pursuing her M.Sc. degree at Tabriz University of Medical Sciences in Medical Genetics.

Mina Jahandideh is pursuing a B.S. degree in Applied Mathematics at Vali-E-Asr University, Rafsanjan, Iran.

Afsaneh Hoseini is pursuing a B.S. degree in Computer Engineering at Islamic Azad University of Arak, Arak, Iran.

Ali Salehzadeh Yazdi graduated from University of Shahid Chamran, Iran, with a B.Sc. degree in Genetics in 2003. He received his M.Sc. degree in Biophysics in 2004 from University of Tehran. He is pursuing his Ph.D. degree in Biophysics.